2.1

$E = \{\text{at least one die is even}\}$

$SUM8 = \{\text{sum of 2 die is 8}\}$

$|\Omega| = 6^2 = 36$

$|SUM8| = 5$ [(2,6), (3,5), (4,4), (5,3), (6,2)]

$P(SUM8) = \frac{5}{36}$

$|SUM8 \cap E| = 3 \rightarrow P(SUM8 \cap E) = \frac{1}{12}$

$P(E|SUM8) = \frac{P(E \cap SUM8)}{P(SUM8)} = \frac{1}{12}\frac{36}{5} = \frac{3}{5}$

Alternatively:

$P(E|SUM8) = \frac{|SUM8 \cap E|}{|SUM8|} = \frac{3}{5}$

2.6

a

$A = \{\text{Alice watches TV}\}$

$B = \{\text{Betty watches TV}\}$

$P(A) = 0.6$

$P(B|A) = 0.8 = \frac{P(AB)}{P(A)}$

$P(AB) = P(B|A) * P(A) = 0.8 * 0.6 = 0.48$

b

$P(B|A^c) = 0$ because the TV is not on

Therefore $P(B) = P(AB) = 0.48$

c

$AB^c = \{\text{Alice watches TV and Betty does not}\}$

$P(B^c|A) = \frac{P(AB^c)}{P(A)} = 1 - P(B|A) = 0.2$

$P(AB^c) = P(B^c|A) * P(A) = 0.2 * 0.6 = 0.12$

2.11

$C = \{\text{customer is careful}\}$

$P(C) = 0.8$

$R = \{\text{customer is reckless}\}$

$P(R) = 0.2$

$A = \{\text{customer caused an acciident}\}$

$P(A|C) = \frac{P(AC)}{P(C)} = 0.01$

$P(A|R) = \frac{P(AR)}{P(R)} = 0.04$

$A = AC \cup AR$

$P(A) = P(AC) + P(AR) - P(AC \cap AR)$

$P(AC) = 0.01 * 0.8 = 0.008$

$P(AR) = 0.04 * 0.2 = 0.008$

$P(AC \cap AR) = 0$ because $C \cap R = \phi$

$P(A) = 0.016$

$P(C|A) = \frac{P(A|C)P(C)}{P(A)} = \frac{0.01*0.8}{0.016} = \frac{1}{2}$

2.14

$P(A \cap B) = 0$ because they are disjoint. So $P(A)P(B) = P(A \cap B)$ for independence. Thus $P(A) = 0 \vee P(B) = 0$ for independence.

2.48

There is a lot of missing information so I will proceed with this question with the following assumptions:

1. There is only a single perpetrator and they live in the town
2. There is a 0% false negative rate with the DNA matching (the perpetrator is guranteed to have their DNA match)

$P(K) = \{\text{Person is the Killer}\}$

$P(M) = \{\text{Person has a DNA Match}\}$

Thus using the above assumptions we get the following values for the probabilities:

$P(K) = \frac{1}{100{,}000}$, obtained from assumption 1.

$P(K^c) = 1 - P(K) = \frac{99{,}999}{100{,}000}$

$P(M|K) = 1$, obtained from assumption 2

$P(M|K^c) = \frac{1}{10{,}000}$, obtained from the problem statements

Thus we are solving for $P(K|M)$, what is the probability someone is the killer given they have a DNA match?

We can use Bayes Formula to solve for $P(K|M)$:

$$P(K|M) = \frac{P(M|K)P(K)}{P(M|K)P(K)+P(M|K^c)P(K^c)}$$

$$P(K|M) = \frac{1 * \frac{1}{100{,}000}}{1 * \frac{1}{100{,}000} + \frac{1}{10{,}000}\frac{99{,}999}{100{,}000}}$$

$$P(K|M) = \frac{10{,}000}{109{,}999}$$

Thus knowing the Kevin has a DNA match means that there is a probability of $\frac{10{,}000}{109{,}999}$ that he is the killer.

2.54

a

Yes it is possible to determine $P(A)$

$P(AB) = \frac{1}{3}P(B)$

$P(AB^c) = \frac{1}{3}(1 - P(B))$

$P(A) = P(AB) + P(AB^c) + P(AB \cap AB^c) = \frac{1}{3}P(B) + \frac{1}{3}(1 - P(B)) + 0 = \frac{1}{3}$

b

Yes A and B are independent because $P(A|B) = \frac{1}{3} = P(A)$